MOTION ENGINEERING · COMPANION SPEC

v1.0 · 28 October 2026



Advanced *Animation* Specification

Coffee Shop Website · Crema

22 production-ready motion specs. Storyboards, keyframes, code architecture, fallbacks and acceptance criteria. Companion to **UI/UX Design Requirements v1.0**.



ANIM-01 STORYBOARD: HERO SCROLL-DRIVEN POUR (SCROLL PROGRESS)

22	5	60fps	100%
animation specs, ANIM-01 → ANIM-22	Must-have motions for launch	target on mid-range phones	specs with a reduced-motion version

DOCUMENT	DATE	OWNER	FOR
Animation Spec v1.0	28 Oct 2026	Tanvir Hasan	Frontend dev team

00 · ABOUT THIS DOCUMENT

How to use this spec

Each animation is specified the same way, so engineers can pick one up and ship it without guesswork. Tokens, palette and brand ("Crema") are inherited from **UI/UX Design Requirements v1.0**; where they differ, that document wins.

Document control

FIELD	VALUE
Title	Coffee Shop Website: Advanced Animation Specification
Version	1.0
Date	28 October 2026
Owner	Tanvir Hasan
Audience	Frontend developers (React / Next.js / TypeScript), motion designers
Parent doc	UI/UX Design Requirements v1.0 (tokens, palette, IDs)
ID scheme	ANIM-01 … ANIM-22; sub-IDs ANIM-xx-AC# for acceptance

Every spec contains

- Concept
- Trigger
- Recommended library
- Code skeleton (TS)
- Mobile / low-power fallback
- Performance budget
- Storyboard key frames
- Timeline / keyframes
- Component architecture
- State model
- Reduced-motion version
- Acceptance criteria

Priority

MUST launch blocker · **SHOULD** expected for v1 · **COULD** delight layer behind a feature flag

> **Note on this PDF:** all illustrations are flat vector renders (no blur effects) so it opens fast on phones. In the product, surfaces use the Liquid Glass treatment from the UI spec.

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Shared engine & rules

All 22 animations share one small runtime: capability tier, tokens, reduced-motion switch, frame budget. Build it first (phase 0).

TIER FULL

Desktop / modern phones: WebGL, Canvas particles, scroll-scrub, 3D, audio sync. Device memory ≥ 4 GB, ≥ 6 cores, no Save-Data.

TIER LITE

Low-power phones, battery saver, Save-Data or < 50 fps measured: SVG/CSS only, particles ÷ 4, no WebGL, scrub replaced by time-based play.

TIER STATIC

`prefers-reduced-motion: reduce` or in-app toggle: no movement beyond ≤ 150 ms opacity fades; final frames shown as static art.

```ts
// lib/motion/tier.ts
export type MotionTier = "full" | "lite" | "static";

export function detectTier(): MotionTier {
  if (typeof window === "undefined") return "lite";
// SSR default
  const mq = (q: string) => window.matchMedia(q).matches;
  if (mq("(prefers-reduced-motion: reduce)")) return
"static";
  if (localStorage.getItem("crema:reduceMotion") === "1")
return "static";
  const nav = navigator as Navigator & { deviceMemory?:
number;
    connection?: { saveData?: boolean } };
  const lowMem = (nav.deviceMemory ?? 8) < 4;
  const lowCpu = (nav.hardwareConcurrency ?? 8) < 6;
  const saveData = nav.connection?.saveData === true;
  return lowMem || lowCpu || saveData ? "lite" : "full";
}

// FPS guard: downgrade at runtime if we drop frames
export function watchFps(onLow: () => void, min = 50) {
  let frames = 0, start = performance.now(), id = 0;
  const tick = (t: number) => {
    frames++;
    if (t - start > 2000) {
      if ((frames * 1000) / (t - start) < min) onLow();
      frames = 0; start = t;
    }
    id = requestAnimationFrame(tick);
  };
  id = requestAnimationFrame(tick);
  return () => cancelAnimationFrame(id);
}
```

```tsx
// app/providers/MotionProvider.tsx
"use client";
import { MotionConfig } from "motion/react";
import { createContext, useContext, useEffect, useState }
from "react";
import { detectTier, watchFps, type MotionTier } from
"@/lib/motion/tier";
import { spring } from "@/lib/motion/tokens";

const TierCtx = createContext<MotionTier>("lite");
export const useTier = () => useContext(TierCtx);

export function MotionProvider({ children }: { children:
React.ReactNode }) {
  const [tier, setTier] = useState<MotionTier>("lite");
  useEffect(() => {
    const t = detectTier(); setTier(t);
    if (t === "full") return watchFps(() =>
setTier("lite"));
  }, []);
  return (
    <TierCtx.Provider value={tier}>
      <MotionConfig reducedMotion="user" transition=
{spring.snappy}>
        {children}
      </MotionConfig>
    </TierCtx.Provider>
  );
}

// lib/motion/tokens.ts  (mirrors UI spec §3)
export const dur = { instant: .1, fast: .16, base: .24,
  medium: .32, slow: .48, slower: .7 } as const;
export const ease = { standard: [.2,0,0,1], outQuart:
[.25,1,.5,1],
  outExpo: [.16,1,.3,1], pour: [.65,0,.35,1] } as const;
export const spring = {
  snappy: { type: "spring", stiffness: 420, damping: 34 },
  gentle: { type: "spring", stiffness: 220, damping: 26 },
  bouncy: { type: "spring", stiffness: 300, damping: 18 },
  liquid: { type: "spring", stiffness: 160, damping: 9,
mass: .8 },
} as const;
```

FOLDER ARCHITECTURE

```
src/
  components/motion/
    hero-pour/      (ANIM-01)
    live-cup/       (ANIM-02, reused by 07, 09, 17)
    transitions/    (ANIM-03)
    fx/             (steam-trail, ripple, particles…)
  lib/motion/
    tier.ts  tokens.ts  springs.ts  useInViewPause.ts
  assets/motion/   *.svg  *.lottie  *.riv  *.glb
```

GLOBAL RULES

- Animate only `transform`, `opacity`, SVG attributes on small paths, or canvas/WebGL.
- Heavy libraries (GSAP, Three.js, Lottie, Rive) are loaded with `next/dynamic` or `import()`, client-only, after first paint.
- Every loop pauses when off-screen (`IntersectionObserver`) or the tab is hidden.
- Long tasks < 50 ms; main-thread scripting per frame ≤ 4 ms.

02 · INDEX

All animations at a glance

Must items are launch blockers. Could items ship behind feature flags (`flags.anim.<id>`) and can be switched off remotely.

ID	ANIMATION	PRIORITY	PRIMARY LIBRARY	TRIGGER
ANIM-01	**Home hero scroll-driven coffee pour**	MUST	GSAP + ScrollTrigger · SVG	Scroll (pinned)
ANIM-02	**Drink customizer live cup**	MUST	Framer Motion · SVG · custom spring	Option change
ANIM-03	**Bean-to-cup page transition**	SHOULD	Framer Motion · Canvas 2D	Route change
ANIM-04	**Add-to-cart flying cup**	MUST	Framer Motion	Tap “Add”
ANIM-05	**Latte-art loader (rosetta / heart)**	MUST	SVG stroke · CSS	Pending > 300 ms
ANIM-06	**Steam cursor / finger trail**	COULD	Canvas 2D	Pointer move
ANIM-07	**3D rotatable cup**	COULD	Three.js · R3F · drei	Drag / idle
ANIM-08	**Magnetic glass buttons + liquid ripple**	SHOULD	Framer Motion · CSS	Hover / press
ANIM-09	**Order-tracking brewing timeline**	MUST	Framer Motion · SVG · SSE	Server event
ANIM-10	**Time-of-day sky & lighting shift**	SHOULD	CSS vars · View Transitions	Clock / toggle
ANIM-11	**Coffee ripple / stain on click**	COULD	CSS · SVG	Click / tap
ANIM-12	**Menu category liquid morph**	SHOULD	Framer Motion · SVG goo	Category tap
ANIM-13	**Loyalty stamp coffee-ring + bean confetti**	SHOULD	Framer Motion · Canvas	Stamp earned
ANIM-14	**Pull-to-refresh pouring drop**	COULD	Framer Motion gestures	Pull gesture
ANIM-15	**Ambient sound sync**	COULD	Web Audio · rAF	Sound toggle
ANIM-16	**Seasonal particle themes**	COULD	Canvas 2D / tsParticles-lite	Season flag
ANIM-17	**Checkout success “cheers”**	SHOULD	Lottie / Rive	Payment success
ANIM-18	**Hold-to-confirm barista tamp**	SHOULD	Framer Motion	Press & hold
ANIM-19	**Origin map bean flight**	COULD	GSAP MotionPath · SVG	Scroll into view
ANIM-20	**Gyro tilt liquid**	COULD	DeviceOrientation · spring	Device tilt
ANIM-21	**Foam-text headline reveal**	COULD	SVG mask · Framer Motion	In view
ANIM-22	**Receipt print-out confirmation**	SHOULD	Framer Motion	Order confirmed

03 · ANIM-01 · MUST

Home hero: scroll-driven coffee pour

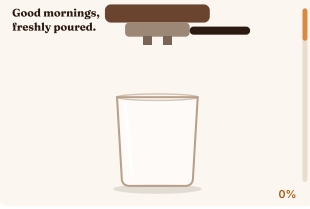
MUST | GSAP 3 + ScrollTrigger | SVG | Framer Motion (handoff) | Optional: Canvas steam

Trigger: **hero enters viewport → scroll progress 0–100 % while pinned**

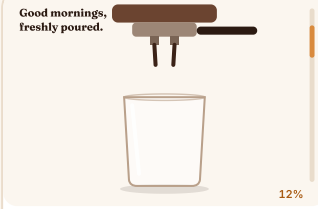
CONCEPT

The hero section pins for **~2.2 viewport heights** of scroll. Scrolling scrubs a pour: an espresso portafilter (or a kettle on the "Pour-over" variant) streams coffee into a glass, the liquid level rises with a living wave surface, a crema layer forms on top, steam rises. Every frame is **fully reversible**: scrolling up un-pours. At 100 % the cup slides aside and the first menu cards slide in, then the section unpins and normal scrolling continues into the menu.

Storyboard (scroll progress shown on the right rail of each frame)



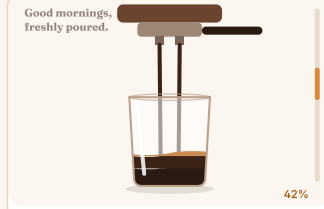
F1 Pinned. Empty glass under portafilter; headline visible.



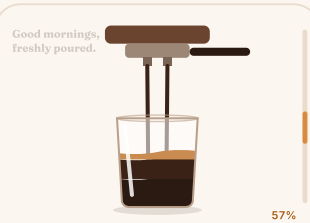
F2 First drops: thin streams extend (scaleY 0→1).



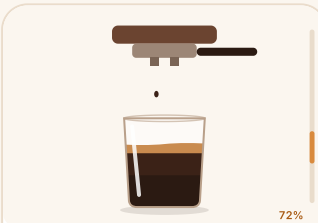
F3 Stream full; level rises, wave surface wobbles.



F4 Crema starts: tan band appears at surface.



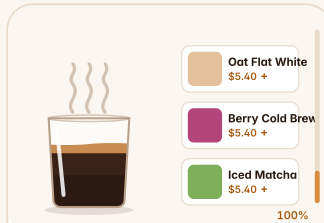
F5 Level 55 %; crema thickens; micro-bubbles.



F6 Stream cuts; a last drip falls.



F7 Steam wisps rise; optional latte-art heart.



F8 Cup slides left; menu cards slide in. Unpin.

VISUAL LAYERS (Z-ORDER)

1 Background mesh (day/night, ANIM-10) · 2 Headline & CTA · 3 Portafilter / kettle SVG · 4 Stream paths ×2 · 5 Cup back wall · 6 Liquid group (coffee, crema, bubbles) clipped to cup · 7 Cup front glass + highlight · 8 Steam (SVG or Canvas) · 9 Menu-card reveal.

VARIANTS

Espresso (default, day) · **Pour-over kettle** (gooseneck arcs a stream; seasonal) · **Night Decaf** (darker palette, slower steam). Variant chosen from the theme; same timeline.

SCROLL DISTANCE

Pin length +=220% desktop, +=160% mobile. `scrub: 0.6` (smoothing 0.6 s). Snap points at 0, 0.58 (full cup) and 1 so users never stop mid-transition.

Pour timeline (scroll-scrubbed)

Positions are scroll progress of the pinned section (0 → 1). Because the timeline is scrubbed, "duration" is distance; easings shape progress within each tween.

PROGRESS	KEYFRAME / PROPERTY	TARGET	EASE
0.00 – 0.05	Pin starts; headline fully visible	—	none (scrub)
0.05 – 0.15	Streams `scaleY 0→1` from spout (transform-origin top)	stream-L, stream-R	power1.in
0.10 – 0.60	Coffee level `y: 0 → -55%` of cup height; wave amplitude 3 px	liquid-group	none (linear w/ scroll)
0.10 – 0.60	Wave phase driven by scroll velocity (`self.getVelocity()`), clamp ±6 px	wave-path	quickTo 0.3 s
0.30 – 0.60	Crema band `scaleY 0→1` , colour #C98B4F → #D9A36A	crema	power2.out
0.15 – 0.55	Headline `opacity 1→0, y 0→-24px`	hero-copy	power1.inOut
0.60 – 0.66	Streams retract (`scaleY 1→0` , origin bottom); last drip falls 40 px	streams, drip	power2.in
0.66 – 0.85	Steam wisps fade in, rise 18 px, stroke-dashoffset loop starts	steam	sine.out
0.80 – 0.86	Latte-art heart draws on surface (optional, FULL tier)	art-path	power1.inOut
0.86 – 1.00	Cup `x: 0 → -32vw` , scale 1→0.8; menu cards stagger in from right (0.04)	cup, menu-cards	power3.inOut
1.00	Unpin; Framer Motion takes over card hover / layoutId to menu	—	—

STATE MODEL

`idle` → `pinned` → `pouring` → `settling` → `steaming` → `revealing` → `released`

Scroll position fully determines state (pure function of progress), so reversing scroll walks backwards through the states. Steam loop & wave idle animation are the only time-based parts and pause when the section is off-screen.

SCROLL ↔ STATE MAPPING

idle < 0 ≤ pinned < .05 ≤ pouring < .60 ≤ settling < .66 ≤ steaming < .86 ≤ revealing < 1 ≤ released. Expose as `data-state` on the stage for CSS hooks & tests.

COMPONENT TREE

```
<HeroPour>                    // client, dynamic(ssr:false)
 ├─ <HeroCopy/>                // SSR text = LCP element
 ├─ <PourStage ref=stage>      // pinned container
 │   ├─ <Portafilter/> | <Kettle/>
 │   ├─ <Streams/>             // 2 paths
 │   ├─ <GlassCup clipId>
 │   │   └─ <Liquid level wave crema/>
 │   └─ <Steam variant=svg|canvas/>
 └─ <MenuTeaser/>              // cards revealed at 0.86
<HeroPourStatic/>             // reduced-motion poster
```

Making the pour safe everywhere

```tsx
// components/motion/hero-pour/HeroPour.tsx
"use client";
import { useRef } from "react";
import gsap from "gsap";
import { ScrollTrigger } from "gsap/ScrollTrigger";
import { useGSAP } from "@gsap/react";
import { useTier } from "@/app/providers/MotionProvider";
gsap.registerPlugin(ScrollTrigger, useGSAP);

export default function HeroPour() {
  const root = useRef<HTMLDivElement>(null);
  const tier = useTier();
  useGSAP(() => {
    if (tier === "static") return;                // poster only
    const mm = gsap.matchMedia();
    mm.add({ desk: "(min-width:768px)", mob: "(max-
width:767px)" },
      (ctx) => {
      const { desk } = ctx.conditions!;
      const tl = gsap.timeline({
        defaults: { ease: "none" },
        scrollTrigger: {
          trigger: root.current, start: "top top",
          end: desk ? "+=220%" : "+=160%",
          pin: true, scrub: 0.6, anticipatePin: 1,
          snap: { snapTo: [0, 0.58, 1], duration: 0.4 },
          onUpdate: (s) => setWave(s.getVelocity()),
        },
      });
      tl.fromTo(".stream", { scaleY: 0 },
             { scaleY: 1, duration: .10 }, .05)
        .to(".liquid", { yPercent: -55, duration: .50 }, .10)
        .fromTo(".crema", { scaleY: 0 },
             { scaleY: 1, ease: "power2.out", duration: .30 },
.30)
        .to(".hero-copy", { autoAlpha: 0, y: -24, duration: .40
}, .15)
        .to(".stream", { scaleY: 0, transformOrigin: "50%
100%",
             ease: "power2.in", duration: .06 }, .60)
        .fromTo(".steam", { autoAlpha: 0, y: 18 },
             { autoAlpha: 1, y: 0, duration: .19 }, .66)
        .to(".cup", { x: "-32vw", scale: .8,
             ease: "power3.inOut", duration: .14 }, .86)
        .from(".menu-card", { x: 80, autoAlpha: 0,
             stagger: .04, duration: .10 }, .88);
    });
    const setWave = gsap.quickTo(".wave", "attr:data-amp",
                                 { duration: .3 });
  }, { scope: root, dependencies: [tier] });
  return <section ref={root} className="hero-pour">{/* … */}
</section>;
}
```



RM STATIC/REDUCED: single poster frame (full cup, steam static).



LITE LITE: time-based 2.4 s play when in view, no pin, no scrub.

MOBILE / LOW-POWER FALLBACK (TIER LITE)

- No pinning on screens < 768 px tall-thin viewports where pin causes address-bar jank on iOS. Instead play the same timeline **once, time-based (2.4 s)** when 40 % in view (`ScrollTrigger toggleActions: "play none none none"`).
- Steam rendered as 3 SVG paths (no Canvas); wave amplitude fixed, no velocity coupling.
- Use `ScrollTrigger.normalizeScroll(true)` only on FULL tier iOS to avoid jitter.

PREFERS-REDUCED-MOTION VERSION (TIER STATIC)

- Render `<HeroPourStatic/>`: the F7 frame as inline SVG (full cup, crema, static steam). No pin, no scrub, no scroll hijack.
- Menu teaser is a normal section below, fading in at 150 ms opacity only.
- GSAP and ScrollTrigger are **not downloaded** in this tier.

Shipping the pour

PERFORMANCE BUDGET

METRIC	BUDGET
JS (gzip)	GSAP core 24 KB + ScrollTrigger 12 KB + component 4 KB ≤ 42 KB, lazy after LCP
SVG	≤ 18 KB inline, ≤ 60 path nodes
Frame cost	≤ 4 ms scripting / frame at 4× CPU throttle; 60 fps desktop, ≥ 50 fps mid Android
LCP impact	0 ms: headline text is SSR LCP; stage hydrates after
CLS	0: stage has fixed aspect-ratio box

TESTING & DEBUGGING

- `?debug=motion` enables ScrollTrigger `markers` and an on-screen progress/state readout.
- Playwright: scroll to fixed offsets, wait 700 ms (scrub settle), screenshot-diff against golden frames F1–F8 (threshold 0.2 %).
- Storybook story with a manual progress slider (0–1) driving `tl.progress()` so designers can review frames without scrolling.
- Lighthouse CI + Web Vitals RUM: alert if hero INP > 200 ms or long tasks > 50 ms appear during scroll.
- Test on iOS Safari (address-bar resize), Android Chrome, Firefox, plus a trackpad with momentum scrolling.

Acceptance criteria

ID	CRITERION
ANIM-01-AC1	Scrubbing from 0 → 100 % and back produces identical frames in reverse (no drift; verified by Playwright screenshots at 0, 25, 58, 85, 100 %).
ANIM-01-AC2	Section pins for exactly the configured distance; no layout jump on unpin (CLS = 0).
ANIM-01-AC3	Snap lands on 0, 0.58 or 1 within 400 ms after the user stops scrolling.
ANIM-01-AC4	Liquid never renders outside the cup (clipPath) at any progress and any breakpoint 360–1920 px.
ANIM-01-AC5	Reduced motion: no pin, no GSAP network request, static poster visible; keyboard users can tab past the hero without scroll traps.
ANIM-01-AC6	60 fps on M1/i5 laptop; ≥ 50 fps on Pixel 6a class phone (Chrome DevTools trace, 10 s scroll).
ANIM-01-AC7	Hero headline LCP < 2.0 s on 4G Moto G profile with animation enabled.

> Kettle variant: same timeline, swap `<Portafilter/>` for `<Kettle/>` whose spout tilts 0 → 28° over 0.05–0.15 and the single stream is a quadratic-bezier path whose control point follows the tilt.

Pour-over kettle variant (seasonal)



K1 Kettle rests above glass.



K2 Spout tilts 14°; thin arc begins.



K3 Full tilt 28°; arc stream; level 50 %.



K4 Kettle returns; bloom + steam.

Drink customizer: the live cup

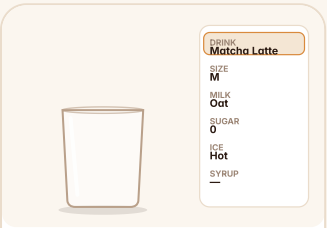
MUST | Framer Motion | SVG layers | Custom spring solver | Optional: R3F (ANIM-07) | Web Audio clinks (ANIM-15)

Trigger: **drink selected / any option change**

CONCEPT

Beside the options panel, a glass **builds the drink in real time**. Selecting a drink pours its base; milk swirls in and blends; foam caps the top with optional latte art. Changing size resizes the cup with a spring; milk type changes the tint; sugar cubes drop with a splash; ice cubes clink and float; syrups drizzle down the inside walls. The liquid surface is a **damped spring**, so every change makes it wobble then settle, which makes the cup feel physical.

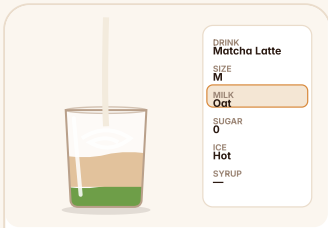
Storyboard: building an oat matcha latte



F1 Select "Matcha Latte": empty glass slides in.



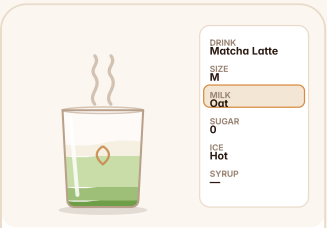
F2 Matcha shot pours: deep green layer, surface wobble.



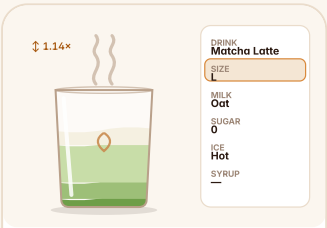
F3 Oat milk pours; white swirl spirals into the green.



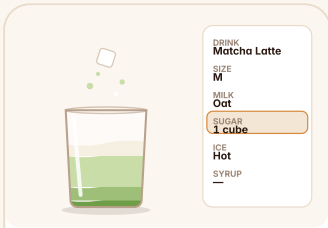
F4 Layers blend (gradient lerp 600 ms) → pale matcha.



F5 Foam cap settles; latte-art leaf draws; steam.



F6 Size L: cup springs to 1.14×; volume re-flows.



F7 Sugar cube drops; splash droplets; ripple.



F8 Iced: cubes clink & float; caramel drizzles walls.

FLAVOUR COLOUR TOKENS (BASE → BLENDED)

DRINK	BASE	BLENDED	NOTE
Matcha latte	#6E9E48	#C8DDA8	Ceremonial green → pale jade after blend
Flat white	#3B2217	#C99467	Ristretto → caramel-brown
Berry cold brew	#4A2A1A	#B4457A	Brew + berry syrup band
Nitro cold brew	#2B1A12	#E8DCCB	Cascade: tan bubbles rise then settle
Chai latte	#A0612F	#E2B98C	Spice flecks (dots) in milk
Hot chocolate	#4E2A1E	#7A4A35	Cocoa dust on foam

MILK TINTS (LERP 35 % INTO BASE)

MILK	TINT
Whole	#F3EBDD
Oat	#E2C29C
Almond	#EAD9C0
Soy	#EFE6CF
Coconut	#F7F4EE
None	transparent

Blend colour = `mix(base, milkTint, milkRatio)` in OKLCH for natural tones.

Option events & keyframes

Every option is an event in a small state machine. The visual cup is a pure function of the recipe state + an animation queue, so rapid changes never desync.

EVENT	VISUAL EFFECT	DURATION	EASING / SPRING
SELECT_DRINK	Cup slides in (x 24→0, opacity); base pours: stream 250 ms then level rises	700 ms	spring.gentle + pour ease
SET_SIZE	Cup scaleX/scaleY to 0.86 / 1 / 1.14; layer heights re-flow to new volume	450 ms	spring {300, 22}
SET_MILK	Milk stream + swirl path (stroke-dashoffset), then colour lerp base → blend	900 ms	pour / standard
SET_TEMP Iced	Steam fades; 3–5 cubes drop (gravity), bob on surface, clink sound	800 ms	spring.bouncy per cube
ADD_SUGAR (n)	Cube falls 60 px, splash droplets (6 particles), ripple ring; level +1 %	550 ms	accelerate → bouncy
ADD_SYRUP	Drizzle paths draw down inner wall, pool tints bottom layer	700 ms	outQuart
SET_FOAM / ART	Foam layer rises; latte-art path draws on top surface	500 + 600 ms	outQuart / pour
ANY change	Surface wave impulse ∝ volume change; damped spring k=160, c=9	~1.2 s settle	spring.liquid

STATE MODEL

`empty` → `pouringBase` → `building` → `settling` → `ready`

Events: SELECT_DRINK, SET_SIZE, SET_MILK, SET_TEMP, ADD_SUGAR, ADD_SYRUP, SET_SHOTS, SET_FOAM. While *building*, new events are queued (max 3, coalesced by type) so taps never feel ignored; the visual target always reflects the latest recipe.

LIQUID SURFACE PHYSICS

Surface = quadratic path with offset *y(t)* from a damped spring: `a = -k·y - c·v`, k = 160, c = 9, mass 0.8. An impulse `v += Δvolume × 40` on each change. Two harmonics (wavelength 1× and 0.5× cup width) give a natural slosh. Settles < 1.2 s; amplitude clamp ±6 px.

```
// lib/cup/types.ts
export type Milk = "whole"|"oat"|"almond"|"soy"|"coconut"|"none";
export interface Recipe {
  drink: DrinkId; size: "S"|"M"|"L"; milk: Milk;
  temp: "hot"|"iced"; sugar: 0|1|2|3|4; shots: number;
  syrups: SyrupId[]; foam: boolean; art:
"none"|"heart"|"leaf"|"rosetta";
}
export interface Layer { id: string; color: string;
  volume: number;            // ml
  blend?: { into: string; ratio: number } }

// pure: recipe → ordered layers (bottom → top)
export function buildLayers(r: Recipe): Layer[] {
  const base = DRINKS[r.drink];
  const vol = SIZES[r.size];            // 240 / 360 / 480
  const layers: Layer[] = [];
  if (r.syrups.length) layers.push(syrupLayer(r.syrups, vol));
  layers.push({ id: "base", color: base.color,
                volume: base.ratio * vol });
  if (r.milk !== "none") layers.push({ id: "milk",
     color: MILK[r.milk], volume: (1 - base.ratio) * vol * .85,
     blend: { into: "base", ratio: .35 } });
  if (r.foam) layers.push({ id: "foam", color: FOAM,
                            volume: vol * .12 });
  return layers;
}
```

```
// lib/cup/useLiquidSpring.ts
import { useAnimationFrame } from "motion/react";
export function useLiquidSpring(k = 160, c = 9, m = .8) {
  const s = useRef({ y: 0, v: 0 });
  const [y, setY] = useState(0);
  const impulse = useCallback((dv: number) => {
    s.current.v += dv * 40;
  }, []);
  useAnimationFrame((_, dt) => {
    const t = Math.min(dt, 32) / 1000;
    const st = s.current;
    const a = (-k * st.y - c * st.v) / m;
    st.v += a * t; st.y += st.v * t;
    st.y = Math.max(-6, Math.min(6, st.y));
    if (Math.abs(st.y) > .01 || Math.abs(st.v) > .01)
      setY(st.y);               // throttled re-render
  });
  return { y, impulse };
}
// wave path for the top surface of a layer
export const wavePath = (w: number, top: number, y: number) =>
  `M0 ${top} Q${w*.25} ${top - y} ${w*.5} ${top}
   T${w} ${top} V200 H0Z`;
```

Building the live cup

```
// components/motion/live-cup/LiveCup.tsx
"use client";
import { motion, AnimatePresence } from "motion/react";
import { buildLayers, type Recipe } from "@/lib/cup";
import { useLiquidSpring, wavePath } from
"@/lib/cup/useLiquidSpring";
import { useTier } from "@/app/providers/MotionProvider";

export function LiveCup({ recipe }: { recipe: Recipe }) {
  const tier = useTier();
  const layers = useMemo(() => buildLayers(recipe), [recipe]);
  const { y, impulse } = useLiquidSpring();
  const scale = { S: .86, M: 1, L: 1.14 }[recipe.size];
  usePrevDiff(layers, (dv) => tier !== "static" && impulse(dv));
  return (
    <motion.svg viewBox="0 0 120 160" aria-hidden
      animate={{ scale }} transition={spring.bouncy}>
      <CupBack />
      <g clipPath="url(#cup-clip)">
        {stack(layers).map((L) => (
          <motion.path key={L.id} fill={L.color}
            initial={false}
            animate={{ d: wavePath(120, L.top, tier==="full" ? y :
0),
                       fill: L.blendColor ?? L.color }}
            transition={{ d: { duration: 0 },
                          fill: { duration: .6 },
                          default: spring.gentle }} />
        ))}
        <AnimatePresence>
          {recipe.temp === "iced" && <IceCubes n={4} />}
          {recipe.sugar > 0 && <SugarDrop key={recipe.sugar} />}
        </AnimatePresence>
        {recipe.art !== "none" && <LatteArt kind={recipe.art} />}
      </g>
      <CupFront /> {recipe.temp === "hot" && <Steam />}
    </motion.svg>
  );
}
// a11y: <p aria-live="polite">{describe(recipe)}</p> rendered by
parent
```

COMPONENT TREE

```
<Customizer>  (form state: useReducer / zustand)
 ├─ <OptionsPanel/>  → dispatch(event)
 ├─ <LiveCup recipe/>
 │   ├─ <CupBack/> <CupFront/>  (static SVG)
 │   ├─ <LiquidLayer/> ×n  (spring surface)
 │   ├─ <Pour stream/>  <MilkSwirl/>
 │   ├─ <IceCubes/> <SugarDrop/> <Drizzle/>
 │   └─ <LatteArt/> <Steam/>
 └─ <PriceBar/>  (digit roll)
```

MOBILE / LOW-POWER (LITE)

- Cup docks to 40 % height at top of sheet; same SVG.
- Surface spring OFF (flat top); colour & height still tween 300 ms.
- Particles (splash, swirl) skipped; ice drops without bob; max 3 cubes.

REDUCED MOTION (STATIC)

- Cup updates instantly to the final recipe state, 150 ms cross-fade.
- No pours, wobble, drops or steam movement; latte art drawn static.
- Text summary in `aria-live` remains the source of truth.

PERFORMANCE BUDGET

METRIC	BUDGET
JS	≤ 9 KB gz (component + spring); Framer Motion shared
DOM	≤ 14 animated SVG nodes at once; particles ≤ 12
Frame	≤ 3 ms / frame; React re-render only cup subtree, spring via rAF
Input	Option tap → first visual change < 100 ms (INP-safe)

ACCEPTANCE CRITERIA

ID	CRITERION
ANIM-02-AC1	Each option change starts its visual response within 100 ms and fully settles within 1.2 s.
ANIM-02-AC2	10 rapid taps in 2 s end in exactly the latest recipe state (no stale layers).
ANIM-02-AC3	Layer heights are proportional to recipe volumes (±2 %) for S/M/L.
ANIM-02-AC4	Milk change visibly re-tints the blended layer to the token colour.
ANIM-02-AC5	Reduced motion: zero animated transforms; summary text announced.
ANIM-02-AC6	≥ 55 fps on Pixel 6a during continuous option changes.

Bean-to-cup page transition

SHOULD | Framer Motion AnimatePresence | Canvas 2D particles | Next.js App Router template.tsx

Trigger: **route change between top-level pages**

CONCEPT

Navigating between main sections plays a short coffee “ritual”: a bean appears, cracks into grind particles that spiral into a cup, the cup fills and the next page rises in. It masks data loading on slow routes and is skipped for back/forward navigations.



`0 ms` Tap link: a single roasted bean scales in at centre.



`220 ms` Bean cracks → 24 grind particles burst out.



`480 ms` Particles spiral down into a cup.



`750 ms` Cup fills; new page rises in behind.

T / PHASE	KEYFRAME	DUR.	EASING
0–220	Bean scale 0→1, rotate −20°→0°	220	outExpo
220–480	Bean opacity → 0; 24 particles burst (radial vel. 120–220 px/s, gravity 600)	260	physics
480–750	Particles attracted to cup mouth; cup level 0→60 %	270	pour
600–900	Next page y 24→0, opacity 0→1; overlay fades	300	outQuart

```
// app/(site)/template.tsx
"use client";
export default function Template({ children }) {
  const path = usePathname();
  const tier = useTier();
  return (
    <AnimatePresence mode="wait" initial={false}>
      <motion.main key={path}
        initial={{ opacity: 0, y: 24 }}
        animate={{ opacity: 1, y: 0,
          transition: { delay: tier==="full" ? .6 : 0 } }}
        exit={{ opacity: 0, transition: { duration: .2 } }}>
        {tier === "full" && <BeanToCupOverlay />}
        {children}
      </motion.main>
    </AnimatePresence>
  );
}
// BeanToCupOverlay: <canvas> particle sim, 24 particles,
// self-unmounts on animationcomplete
```

STATE MODEL

`idle` → `exiting` → `grinding` → `brewing` → `entering`

Skipped if route data resolves < 120 ms and user prefers speed (setting).

MOBILE / LOW-POWER

LITE: no particles; bean → cup icon cross-fade (300 ms) then page fade. Never blocks navigation > 400 ms.

REDUCED MOTION

STATIC: plain 150 ms cross-fade between pages; no overlay rendered.

PERFORMANCE BUDGET

≤ 900 ms total; canvas ≤ 24 particles; overlay JS ≤ 4 KB gz; zero CLS (overlay is fixed).

ACCEPTANCE CRITERIA

`ANIM-03-AC1` Total transition ≤ 900 ms (FULL) / 400 ms (LITE); never delays interactivity of the new page.

`ANIM-03-AC2` Back/forward navigation skips the overlay.

`ANIM-03-AC3` Focus moves to the new page <h1> after transition; screen readers announce the route.

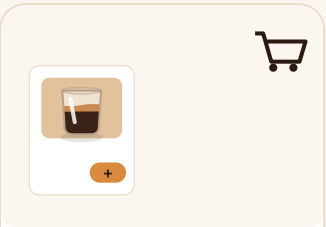
05 · ANIM-04 · MUST

Add-to-cart flying cup

MUST | Framer Motion | motion values + useAnimate | Portal overlay | Trigger: **tap "+" / "Add to order"**

CONCEPT

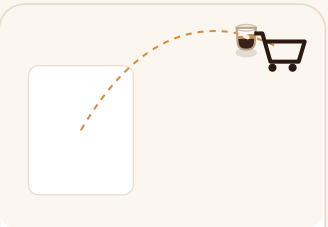
A miniature of the configured drink jumps out of the card and flies along a curved arc into the cart icon, which bounces and pops its badge. Gives unmistakable confirmation and teaches where the cart lives.



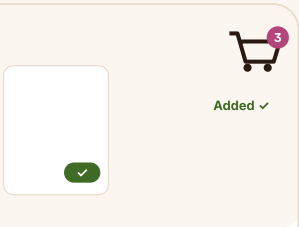
0 Press "+": button scales .92; cup lifts.



180 ms Cup clone flies along bezier arc.



420 ms Approaches cart, shrinks to 0.4, rotates 12°.



600 ms Cart bounces; badge pops 1→1.35→1.

T / PHASE	KEYFRAME	DUR.	EASING
0–100	Button scale .92 → 1; cup clone created at card image rect	100	snappy
100–520	Clone follows quadratic bezier (control = midpoint − 30 % viewport height); scale 1 → .4; rotate 0→12°	420	outQuart on t
520–600	Clone fades; cart y 0 → −4 → 0	180	bouncy
520–760	Badge scale 1 → 1.35 → 1; count ticks	240	bouncy

```
export function useFlyToCart() {
  const [scope, animate] = useAnimate();
  return async (from: DOMRect, to: DOMRect) => {
    const el = spawnClone(from);            // portal <div>
    const cx = (from.x + to.x) / 2;
    const cy = Math.min(from.y, to.y) - innerHeight * .3;
    const t = motionValue(0);
    t.on("change", (v) => {
      const x = qBez(from.x, cx, to.x, v);
      const y = qBez(from.y, cy, to.y, v);
      el.style.transform =
        `translate(${x}px,${y}px) scale(${1-.6*v})
rotate(${12*v}deg)`;
    });
    await animate(t, 1, { duration: .42, ease: [.25,1,.5,1] });
    el.remove();
    animate("#cart", { y: [0, -4, 0] }, spring.bouncy);
    animate("#cart-badge", { scale: [1, 1.35, 1] },
spring.bouncy);
  };
}
```

STATE MODEL

`idle` → `flying` → `landing` → `done`

Multiple adds run concurrently (max 3 clones); extra adds just bump the badge.

MOBILE / LOW-POWER

LITE: no clone; button morphs to "Added ✓" and badge pops. Same timing.

REDUCED MOTION

STATIC: badge count updates; toast "Added to order" (no movement).

PERFORMANCE BUDGET

≤ 760 ms; 1 composited layer per clone; no layout reads during flight (rects read once).

ACCEPTANCE CRITERIA

`ANIM-04-AC1` Clone path ends exactly on cart icon centre at every breakpoint and with scrolled page.

`ANIM-04-AC2` Cart count updates even if animation is interrupted by navigation.

`ANIM-04-AC3` Focus remains on the triggering button; live region announces "Added Oat Flat White".

05 · ANIM-05 · MUST

Latte-art loader

MUST | SVG stroke-dashoffset | CSS @keyframes | Optional Lottie/Rive

Trigger: **any pending state > 300 ms**

CONCEPT

Replaces every generic spinner. Inside a crema-coloured disc, milk-foam strokes draw a heart (short waits) or a rosetta (longer waits), loop with a soft pulse. Pure CSS/SVG so it works before JS hydrates.



`0` Crema disc fades in (scale .9 → 1).



`400 ms` Heart path draws (dashoffset).



`800 ms` Heart complete; gentle pulse.



`1.2 s loop` Variant: rosetta leaves stack, stem pulls through.

T / PHASE	KEYFRAME	DUR.	EASING
0–150	Disc opacity 0→1, scale .9→1	150	outQuart
150–800	Heart dashoffset L → 0	650	pour
800–1000	Pulse scale 1 → 1.04 → 1	200	sine
1000–1200	Fade strokes, restart (rosetta after 3 loops)	200	standard

```
// components/motion/LatteLoader.tsx (server-safe)
export function LatteLoader({ label = "Loading" }) {
  return (
    <span role="status" className="latte">
      <svg viewBox="0 0 80 80" aria-hidden>
        <circle cx="40" cy="40" r="36" className="crema" />
        <path pathLength={1} className="art"
          d="M40 58C24 48 22 32 32 29c5-1 8 3 8 6 0-3 3-7 8-6
             10 3 8 19-8 29z" />
      </svg>
      <span className="sr-only">{label}…</span>
    </span>
  );
}
/* latte.css */
.art { stroke-dasharray:1; stroke-dashoffset:1;
  animation: draw 1.2s var(--ease-pour) infinite; }
@keyframes draw { 65%,100% { stroke-dashoffset:0 } }
@media (prefers-reduced-motion: reduce) {
  .art { animation:none; stroke-dashoffset:0 } }
```

STATE MODEL

`hidden` → `delay300` → `drawing` → `looping` → `done`

Shown only after 300 ms to avoid flashes; min visible time 500 ms once shown.

MOBILE / LOW-POWER

Identical (pure CSS). On very low-end devices the pulse is dropped.

REDUCED MOTION

Static completed heart + "Loading…" text; no looping.

PERFORMANCE BUDGET

≤ 1.5 KB SVG+CSS; compositor-only except stroke (tiny path); no JS.

ACCEPTANCE CRITERIA

`ANIM-05-AC1` Never visible for waits < 300 ms; once shown stays ≥ 500 ms.

`ANIM-05-AC2` role=status with text label; works with JS disabled.

`ANIM-05-AC3` Heart and rosetta variants both loop seamlessly at 1.2 s.

Steam cursor / finger trail

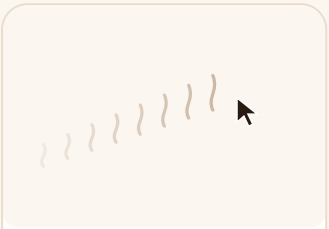
COULD · Canvas 2D · simplex-noise (2 KB) · Pointer Events · Trigger: **pointer move over hero & empty states**

CONCEPT

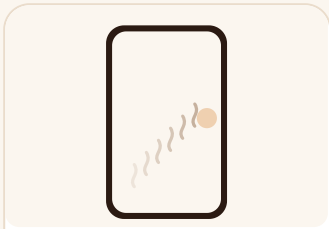
Over the hero and coffee illustrations, the cursor (or finger) leaves gentle steam wisps that rise, curl and dissolve, as if you were moving through the warm air above a cup. Limited to decorative zones so it never interferes with reading.



`0` Pointer moves: wisps spawn at 30 px intervals.



`300 ms` Trail curls upward with curl-noise drift.



`touch` Touch: finger drag leaves a soft trail.

`+600 ms` Idle 600 ms: wisps fade & rise out.

T / PHASE	KEYFRAME	DUR.	EASING
spawn	New particle every 30 px of pointer travel (max 40 alive)	—	—
0–1.6 s	Particle rises 40 px, x drift by simplex noise, scale 0.6→1.4	1600	linear
0–1.6 s	Opacity 0 → .35 → 0	1600	sine in-out
idle	Emitter stops 100 ms after last move	—	—

```
export function SteamTrail({ zone }: { zone:
RefObject<HTMLElement> }) {
  const cv = useRef<HTMLCanvasElement>(null);
  useEffect(() => {
    const ctx = cv.current!.getContext("2d")!;
    const ps: P[] = []; let last = {x:0,y:0}, raf = 0;
    const onMove = (e: PointerEvent) => {
      if (dist(last, e) < 30 || ps.length > 40) return;
      ps.push(spawn(e.offsetX, e.offsetY)); last = e;
    };
    const loop = (t: number) => {
      ctx.clearRect(0, 0, W, H);
      for (const p of ps) step(p, noise2D, t) && drawWisp(ctx,
p);
      prune(ps); raf = requestAnimationFrame(loop);
    };
    zone.current!.addEventListener("pointermove", onMove,
                                   { passive: true });
    raf = requestAnimationFrame(loop);
    return () => { cancelAnimationFrame(raf); /* remove */ };
  }, []);
  return <canvas ref={cv} className="pointer-events-none" />;
}
```

STATE MODEL

`off` → `emitting` → `decaying` → `off`

Canvas is pointer-events:none; one per decorative zone.

MOBILE / LOW-POWER

Disabled on touch for LITE (battery); FULL touch caps at 20 particles.

REDUCED MOTION

Not rendered at all.

PERFORMANCE BUDGET

≤ 40 particles, ≤ 1.5 ms/frame; canvas DPR capped at 1.5; loop stops when no particles.

ACCEPTANCE CRITERIA

`ANIM-06-AC1` Loop is fully idle (no rAF) when there are no live particles.

`ANIM-06-AC2` Never covers text or controls (pointer-events none, zone-limited).

`ANIM-06-AC3` Disabled automatically when FPS guard downgrades to LITE.

3D rotatable cup

COULD | Three.js | React Three Fiber | drei (PresentationControls, Environment) | Draco glTF

Trigger: **customizer / product page: drag or idle**

CONCEPT

An optional 3D glass cup with the user's exact layers (shared data from ANIM-02) that can be spun by drag/swipe, with physically based glass, refraction and soft studio lighting. Used as a 'preview' toggle in the customizer and on seasonal product pages.



`idle` Idle: cup floats ±4 px, slow 8° sway.



`drag` Drag: yaw follows pointer (1 px = 0.6°).



`release` Release: inertia spin, damped.



`settle` Settles to front; label & layers match recipe.

T / PHASE	KEYFRAME	DUR.	EASING
idle	Float y ±4 px, yaw ±8°	4000 loop	sine
drag	Yaw = Δx × 0.6°, pitch clamp ±15°	live	—
release	Angular velocity decays (damping 4)	~900	spring
layers	Liquid uniforms lerp to recipe colours	600	standard

```
const Cup3D = dynamic(() => import("./Cup3DScene"), {
  ssr: false, loading: () => <img src="/cup-poster.avif" alt=""
/>,
});
// Cup3DScene.tsx
export default function Cup3DScene({ recipe }: { recipe: Recipe
}) {
  const { nodes } = useGLTF("/models/cup.glb", true); // draco
  return (
    <Canvas dpr={[1, 1.5]} frameloop="demand"
            camera={{ fov: 30, position: [0, 0.4, 3] }}>
      <Environment preset="apartment" />
      <PresentationControls global snap polar={[-.25, .25]}
          azimuth={[-Infinity, Infinity]} config={{ mass: 1,
tension: 170 }}>
        <Float speed={1.2} floatIntensity={.4}>
          <mesh geometry={nodes.Glass.geometry}>
            <MeshTransmissionMaterial thickness={.3} roughness=
{.05} />
          </mesh>
          <LiquidStack layers={buildLayers(recipe)} />
        </Float>
      </PresentationControls>
    </Canvas>
  );
}
```

STATE MODEL

`poster` → `loading` → `interactive` → `dragging` → `settling`

Poster AVIF shown until model + env map are loaded on interaction or idle.

MOBILE / LOW-POWER

LITE: never loads WebGL; shows rotating 8-frame sprite (AVIF, 60 KB) on drag.

REDUCED MOTION

Static poster image; drag disabled; no float.

PERFORMANCE BUDGET

glb ≤ 500 KB (Draco), env ≤ 150 KB; three+R3F ≤ 250 KB gz loaded on demand; frameloop=demand; 60 fps desktop.

ACCEPTANCE CRITERIA

`ANIM-07-AC1` Model loads only after user taps "3D preview" or idle on FULL tier.

`ANIM-07-AC2` Layer colours match ANIM-02 within ΔE < 3.

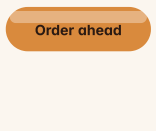
`ANIM-07-AC3` WebGL context lost → poster fallback without errors.

Magnetic glass buttons + liquid ripple

SHOULD | Framer Motion (useMotionValue, useSpring) | CSS radial-gradient / SVG clip | Trigger: **pointer within 80 px; press**

CONCEPT

Primary CTAs feel like a glass bead of caramel: within 80 px they are gently pulled toward the cursor; the specular sheen tracks the pointer; pressing sends a liquid ripple from the exact touch point; releasing makes the capsule wobble like jelly before settling.



`rest` Rest: cursor outside 80 px radius.



`hover` Magnet: button drifts ≤ 8 px toward cursor; sheen follows.



`press` Press: liquid ripple expands from touch point.



`release` Release: jelly wobble, settles via spring.

T / PHASE	KEYFRAME	DUR.	EASING
hover	x,y = (pointer − centre) × 0.25, clamp ±8 px	spring	{350, 20}
press	scale .96; ripple r 0→1.4×width, opacity .5→0	500	outQuart
release	scaleX 1.04 / scaleY .96 → 1	~400	spring {500, 12}
leave	x,y → 0	spring	{300, 24}

```
export function MagneticButton(props: ButtonProps) {
  const ref = useRef<HTMLButtonElement>(null);
  const x = useSpring(0, { stiffness: 350, damping: 20 });
  const y = useSpring(0, { stiffness: 350, damping: 20 });
  const [ripples, add] = useRipples();
  const onMove = (e: React.PointerEvent) => {
    const r = ref.current!.getBoundingClientRect();
    x.set(clamp((e.clientX - (r.left + r.width/2)) * .25, -8,
8));
    y.set(clamp((e.clientY - (r.top + r.height/2)) * .25, -8,
8));
    ref.current!.style.setProperty("--mx", `${e.clientX -
r.left}px`);
  };
  return (
    <motion.button ref={ref} style={{ x, y }} {...props}
      onPointerMove={onMove}
      onPointerLeave={() => { x.set(0); y.set(0); }}
      onPointerDown={(e) => add(e)}
      whileTap={{ scale: .96 }}
      transition={{ type: "spring", stiffness: 500, damping: 12
}}>
      {ripples}{props.children}
    </motion.button>
  );
}
```

STATE MODEL

`rest` → `attracted` → `pressed` → `wobbling` → `rest`

Only on (hover:hover) and (pointer:fine) for magnet; ripple works on touch.

MOBILE / LOW-POWER

LITE/touch: no magnet; ripple + press scale only.

REDUCED MOTION

No magnet, no wobble; press = colour darken 100 ms; focus ring unchanged.

PERFORMANCE BUDGET

No layout reads per frame beyond 1 rect on pointerenter; ripples ≤ 2 concurrent; transform-only.

ACCEPTANCE CRITERIA

`ANIM-08-AC1` Magnet never moves the hit-area more than 8 px; click target still ≥ 44 px.

`ANIM-08-AC2` Ripple originates at pointer/touch coordinates (±2 px).

`ANIM-08-AC3` Keyboard activation (Enter/Space) shows centred ripple.

Order-tracking brewing timeline

MUST | Framer Motion | SVG ring + LiveCup (ANIM-02) | SSE / WebSocket | Lottie (ready burst)

Trigger: **server status event**

CONCEPT

The tracking page shows a progress ring wrapped around the customer's own drink. Each real kitchen stage (Received → Grinding → Brewing → Steaming → Ready) animates the ring and fills the cup further; steam intensifies near the end and a bean-burst celebrates Ready.



stage 1 Received: empty cup, ring starts.



stage 2 Grinding: ring 35 %, grinder shake.



stage 3 Brewing: espresso layer pours.



stage 5 Ready: ring completes, bean burst.

T / PHASE	KEYFRAME	DUR.	EASING
on event	Ring dashoffset to stage % (0/.2/.35/.66/.85/1)	800	pour
on event	Cup level + layer for that stage (reuse ANIM-02)	600	spring.gentle
active	Shimmer sweep along active arc	1600 loop	linear
Ready	Bean burst Lottie + code scale .8→1	1200	outExpo

```
export function BrewTracker({ orderId }: { orderId: string }) {
  const status = useOrderStream(orderId);      // SSE, poll fallback
  const p = STAGE_PROGRESS[status.stage];       // 0..1
  const L = 2 * Math.PI * 34;
  return (
    <figure aria-labelledby="trk">
      <svg viewBox="0 0 124 124">
        <circle className="track" r={34} cx={62} cy={62} />
        <motion.circle r={34} cx={62} cy={62}
          strokeDasharray={L}
          animate={{ strokeDashoffset: L * (1 - p) }}
          transition={{ duration: .8, ease: [.65,0,.35,1] }} />
        <LiveCup recipe={status.recipe} fill={p} compact />
      </svg>
      <p id="trk" aria-live="polite">{stageLabel(status)}</p>
      {status.stage === "ready" && <ReadyBurst />}
    </figure>
  );
}
```

STATE MODEL

`received` → `grinding` → `brewing` → `steaming` → `ready` → `collected`

Driven only by server events; never advances on a client timer.

MOBILE / LOW-POWER

LITE: ring + cup fill only; no shimmer, no Lottie (static check badge).

REDUCED MOTION

Instant step changes; text + icon per stage; no burst.

PERFORMANCE BUDGET

SSE connection only while page visible; Lottie ≤ 60 KB lazy; ≤ 2 ms/frame.

ACCEPTANCE CRITERIA

`ANIM-09-AC1` UI reflects a new stage ≤ 3 s after server event (≤ 15 s on poll fallback).

`ANIM-09-AC2` Stage never regresses visually unless server sends a correction.

`ANIM-09-AC3` Each stage is announced via aria-live and shown with icon + text.

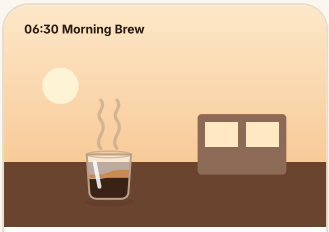
Time-of-day sky & lighting shift

SHOULD | CSS custom properties | View Transitions API | Framer Motion (sun path)

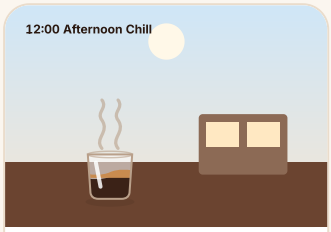
Trigger: **local clock (checked each 5 min) or manual theme toggle**

CONCEPT

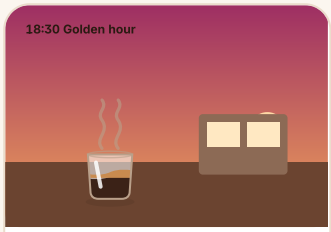
The hero ambience follows the visitor's local time: background sky gradient, sun/moon position, window light and shadow direction move continuously through Morning Brew, Afternoon Chill, Golden hour and Night Decaf. Theme tokens switch at the boundaries with a cross-fade.



06:00 Dawn: warm crema sky, low sun.



12:00 Noon: high sun, cool matcha accents.



18:00 Sunset: berry → caramel gradient.



22:00 Night: moon, lit windows, Night Decaf.

T / PHASE	KEYFRAME	DUR.	EASING
continuous	--sky-top / --sky-bottom interpolated (OKLCH) by minute of day	5-min steps	linear
continuous	Sun/moon follows arc path (offset-path) by time	—	—
boundary	Theme tokens swap via document.startViewTransition	600	standard
manual toggle	Instant jump with 600 ms cross-fade	600	standard

```
// lib/theme/timeOfDay.ts
const STOPS = [ { h: 6,  top: "#FDE7C8", bot: "#F6BE84", theme:
"morning" },
  { h: 12, top: "#CFE6F7", bot: "#F4E6D4", theme: "afternoon"
},
  { h: 18, top: "#9E2F63", bot: "#F2A65A", theme: "golden" },
  { h: 21, top: "#140E0B", bot: "#2A2560", theme: "night" } ];

export function applySky(d = new Date()) {
  const m = d.getHours() * 60 + d.getMinutes();
  const { a, b, t } = bracket(STOPS, m / 60);
  const root = document.documentElement.style;
  root.setProperty("--sky-top", mixOklch(a.top, b.top, t));
  root.setProperty("--sky-bot", mixOklch(a.bot, b.bot, t));
  root.setProperty("--sun-progress", String(m / 1440));
  const theme = (t < .5 ? a : b).theme;
  if (document.documentElement.dataset.tod !== theme) {
    const go = () => (document.documentElement.dataset.tod =
theme);
    document.startViewTransition ?
document.startViewTransition(go) : go();
  }
}
// inline <script> in <head> runs applySky() before paint
```

STATE MODEL

`morning` → `afternoon` → `golden` → `night`

Manual override persists (localStorage) and stops auto-shift until reset.

MOBILE / LOW-POWER

LITE: updates only at theme boundaries (no per-5-min interpolation).

REDUCED MOTION

Theme still follows time, but swaps instantly; sun static.

PERFORMANCE BUDGET

Pure CSS vars; ≤ 1 KB JS; runs on 5-min interval only while tab visible.

ACCEPTANCE CRITERIA

`ANIM-10-AC1` Correct theme applied before first paint (no flash) based on local time.

`ANIM-10-AC2` All text keeps WCAG AA contrast at every interpolated sky colour.

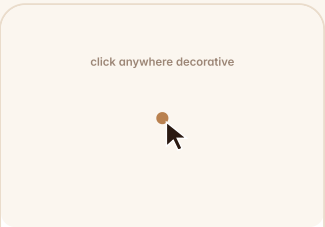
`ANIM-10-AC3` Manual toggle overrides auto mode and persists across visits.

Coffee ripple / stain on click

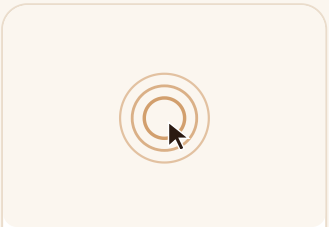
COULD · CSS @keyframes · SVG · Pointer Events · Trigger: **click / tap on non-interactive background areas**

CONCEPT

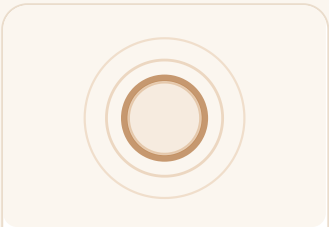
A playful Easter egg: clicking empty background areas (hero, footer, about) drops a bead of coffee that ripples outward and leaves a soft coffee-ring stain that slowly fades, like a cup set down on a café table.



`0` Click: a drop lands.



`200 ms` Ripples expand like a tapped latte.



`500 ms` A coffee-ring stain forms.

`2.5 s` Stain fades out completely.

T / PHASE	KEYFRAME	DUR.	EASING
0–150	Drop scale 0→1 at pointer	150	outExpo
100–700	3 rings r 0→36 px staggered 80 ms, opacity .9→0	600	outQuart
400–900	Stain ring draws (scale .9→1, opacity 0→.8)	500	outQuart
900–2500	Stain fades to 0, element removed	1600	standard

```
export function useCoffeeRipple(zone: RefObject<HTMLElement>) {
  useEffect(() => {
    const el = zone.current!;
    const onClick = (e: MouseEvent) => {
      if ((e.target as Element).closest("a,button,input,[role]"))
return;
      if (getSelection()?.toString()) return;
      if (el.querySelectorAll(".ripple").length ≥ 3) return;
      const r = document.createElement("span");
      r.className = "ripple";
      r.style.cssText = `left:${e.offsetX}px;top:${e.offsetY}px`;
      r.addEventListener("animationend", () => r.remove(),
                         { once: true });
      el.appendChild(r);
    };
    el.addEventListener("click", onClick);
    return () => el.removeEventListener("click", onClick);
  }, [zone]);
}
/* .ripple::before/::after = rings; .ripple = stain;
   animation: ring .6s var(--ease-out-quart), stain 2.5s */
```

STATE MODEL

`idle` → `ripple` → `stain` → `fade` → `removed`

Max 3 concurrent; ignored on links, buttons, inputs, text selection.

MOBILE / LOW-POWER

Same (CSS only); max 1 concurrent.

REDUCED MOTION

Disabled.

PERFORMANCE BUDGET

CSS only, ≤ 0.6 KB; 3 elements max; transform/opacity only.

ACCEPTANCE CRITERIA

`ANIM-11-AC1` Never triggers on interactive elements or while selecting text.

`ANIM-11-AC2` DOM nodes are removed after the animation (no leaks).

`ANIM-11-AC3` No effect when reduced motion is on.

Menu category liquid morph

SHOULD | Framer Motion layoutId | SVG gooey filter (feGaussianBlur + feColorMatrix) | CSS color-mix

Trigger: **category chip tap / scroll-spy change**

CONCEPT

The active-category indicator behaves like a drop of the category's drink: it stretches toward the new chip, overshoots with a little drip, and changes colour to that category (espresso brown → cold-brew dark → matcha green → bakery caramel). The grid below cross-fades in sync.



0 Espresso active (caramel-brown blob).



140 ms Blob stretches like liquid toward target.



300 ms Lands on Matcha, overshoot + drip; colour → matcha.



460 ms Settled; grid cross-fades with 30 ms stagger.

T / PHASE	KEYFRAME	DUR.	EASING
0–160	Leading edge springs to target; trailing edge lags (2 springs)	160	{600, 30} / {300, 30}
160–320	Trailing edge catches up; height squash 1.1→1	160	spring
0–300	Fill colour lerp to category token	300	standard
300–460	Drip ellipse falls 6 px & fades; grid stagger 30 ms	160	outQuart

STATE MODEL

`idle` → `stretching` → `landing` → `settled`

Scroll-spy changes reuse the same animation at 0.6× intensity.

```
export function LiquidTabs({ items, active, onChange }:
TabsProps) {
  const ref = useRef<HTMLDivElement>(null);
  const rect = useChipRect(ref, active);            // {x, w}
  const lead  = useSpring(rect.x, { stiffness: 600, damping: 30
});
  const trail = useSpring(rect.x, { stiffness: 300, damping: 30
});
  useEffect(() => { lead.set(rect.x); trail.set(rect.x); },
[rect.x]);
  const left  = useTransform([lead, trail], ([a, b]) =>
Math.min(a, b));
  const width = useTransform([lead, trail],
                ([a, b]) => Math.abs(a - b) + rect.w);
  return (
    <div ref={ref} role="tablist" className="relative">
      <svg className="absolute inset-0" style={{ filter:
"url(#goo)" }}>
        <motion.rect x={left} width={width} height={36} rx={18}
          animate={{ fill: CATEGORY_COLOR[active] }} />
      </svg>
      {items.map((it) => (
        <button role="tab" aria-selected={it.id === active}
                key={it.id} onClick={() => onChange(it.id)}>
{it.label}</button>
      ))}
    </div>
  );
}
```

MOBILE / LOW-POWER

LITE: single spring pill (no goo filter, no drip).

REDUCED MOTION

Indicator jumps; colour changes instantly; grid fade 150 ms.

PERFORMANCE BUDGET

SVG filter only on a 44 px-tall strip; ≤ 1 ms/frame; no reflow of grid during morph.

ACCEPTANCE CRITERIA

`ANIM-12-AC1` Indicator lands exactly under the active chip at all widths (±1 px).

`ANIM-12-AC2` Text on the coloured blob keeps ≥ 4.5:1 for every category colour.

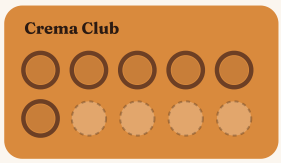
`ANIM-12-AC3` Tabs are a valid ARIA tablist with arrow-key navigation.

Loyalty stamp: coffee-ring + bean confetti

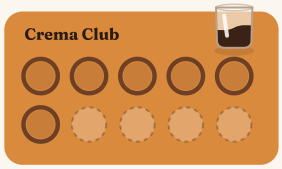
SHOULD | Framer Motion | Canvas 2D (confetti) | Optional Rive (stamp) | Trigger: **server confirms a stamp earned**

CONCEPT

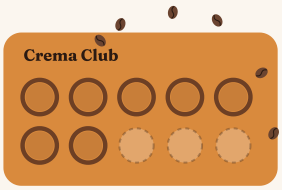
Earning a stamp feels like setting a hot cup down on the card: a cup silhouette lowers onto the next slot, squashes, lifts away leaving an imperfect coffee-ring stamp, and a burst of 12 roasted-bean confetti pops out. Card tilts slightly in 3D.



`0` Card shows 6/10; next slot glows.



`250 ms` Cup descends onto slot 7 (squash on contact).



`450 ms` Coffee-ring stamp appears; beans burst.



`1.2 s` Settle; progress copy updates.

T / PHASE	KEYFRAME	DUR.	EASING
0–250	Cup y −40 → 0, scale 1 → 1.04/0.96 on contact	250	accelerate
250–450	Cup lifts & fades; ring stroke draws (random 6° gap)	200	outQuart
300–1100	12 bean particles: v 180–260 px/s, gravity 900, spin	800	physics
0–600	Card rotateX 0→6→0	600	spring.bouncy

```
export function StampSlot({ filled, justEarned }: Props) {
  const tier = useTier();
  return (
    <div className="slot">
      <AnimatePresence>
        {justEarned && tier === "static" && (
          <motion.div className="cup-press" key="press"
            initial={{ y: -40, opacity: 0 }}
            animate={{ y: [ -40, 0, -24 ], opacity: [0, 1, 0],
                       scaleY: [1, .96, 1] }}
            transition={{ duration: .45, times: [0, .55, 1] }} />
        )}
      </AnimatePresence>
      {filled && (
        <motion.svg className="ring" initial={justEarned &&
"hidden"}
          animate="shown" variants={ringVariants}>
          <motion.circle r={17} cx={20} cy={20}
            pathLength={1} strokeDasharray="0.94 0.06" />
        </motion.svg>
      )}
      {justEarned && tier === "full" && <BeanConfetti count={12}
/>}
    </div>
  );
}
```

STATE MODEL

`idle` → `pressing` → `stamped` → `celebrating` → `idle`

10th stamp: card flips to “Free drink unlocked” (rotateY 180°).

MOBILE / LOW-POWER

LITE: ring draw + 4 CSS bean particles; no card tilt.

REDUCED MOTION

Ring appears with 150 ms fade; text update only.

PERFORMANCE BUDGET

Confetti canvas ≤ 12 particles, auto-unmount ≤ 1.2 s; ≤ 2 ms/frame.

ACCEPTANCE CRITERIA

`ANIM-13-AC1` Stamp animation only fires after server confirmation (no optimistic fake).

`ANIM-13-AC2` Each stamp ring looks unique (seeded rotation & gap).

`ANIM-13-AC3` Announces “Stamp 7 of 10 earned” via aria-live.

Pull-to-refresh pouring drop

COULD | Framer Motion drag / useMotionValue | SVG | Trigger: **pull gesture on Orders & Rewards lists (mobile)**

CONCEPT

On mobile list screens, pulling down reveals a tiny portafilter whose drop of espresso grows with the pull distance; past the threshold a cup appears, the drop falls in with a splash, and the list refreshes while the cup fills as a progress indicator.



`pull` Pull 30 %: drop forms at spout.



`pull` Pull 70 %: cup appears; drop stretches.



`release` Release ≥ threshold: drop falls, splash.



`done` Refreshed list springs back up.

T / PHASE	KEYFRAME	DUR.	EASING
0–72 px	Drop scale = pull/72; list y = pull × 0.5 (rubber band)	live	—
≥ 72 px	Haptic tick; cup fades in	120	outQuart
release	Drop falls 30 px; splash 2 particles	260	accelerate
loading	Cup fill = fetch progress; list returns	spring	{400, 32}

```
export function PourRefresh({ onRefresh, children }: Props) {
  const y = useMotionValue(0);
  const drop = useTransform(y, [0, 72], [0, 1], { clamp: true });
  const [state, set] = useState<"idle"|"armed"|"refreshing">
("idle");
  return (
    <div style={{ overscrollBehaviorY: "contain" }}>
      <PourHeader drop={drop} state={state} />
      <motion.div drag="y" dragConstraints={{ top: 0, bottom: 0
}}
        dragElastic={{ top: 0, bottom: .5 }} style={{ y }}
        onDrag={(_, i) => set(i.offset.y > 72 ? "armed" :
"idle")}
        onDragEnd={async (_, i) => {
          if (i.offset.y > 72) { set("refreshing"); await
onRefresh(); }
          set("idle");
        }}>
        {children}
      </motion.div>
    </div>
  );
}
```

STATE MODEL

`idle` → `pulling` → `armed` → `refreshing` → `done`

Only when list scrollTop = 0; native overscroll disabled via overscroll-behavior.

MOBILE / LOW-POWER

Same gesture; splash skipped; cup static icon.

REDUCED MOTION

Visible “Refresh” button instead of gesture animation; latte loader while loading.

PERFORMANCE BUDGET

Gesture handlers passive where possible; ≤ 1 ms/frame; no layout during drag.

ACCEPTANCE CRITERIA

`ANIM-14-AC1` Refresh triggers only past 72 px and only from scroll-top.

`ANIM-14-AC2` A visible Refresh button exists for keyboard & screen-reader users.

`ANIM-14-AC3` Releasing below threshold springs back without refreshing.

Ambient sound sync

COULD | Web Audio API (AnalyserNode) | rAF | Howler-free (native) | Trigger: **user toggles “Café sound” in the nav**

CONCEPT

An opt-in ambience: espresso-machine hiss, soft chatter and rain on the window, mixed by time of day. Visuals breathe with the sound: hero steam height and sway, the nav speaker icon’s waves, and the stamp ‘thunk’ are driven by the live audio level.



`off` Default: sound OFF (never autoplays).



`on` User enables: soft café loop fades in 1.5 s.



`sync` Steam height & sway follow audio RMS.



`pause` Tab hidden / 30 s idle: fade out, pause.

T / PHASE	KEYFRAME	DUR.	EASING
toggle on	Gain 0 → 0.35 (user volume)	1500	linear ramp
each frame	RMS (AnalyserNode, 256 fft) → `--audio-level` 0..1, smoothed	rAF	lerp .15
events	One-shots: grinder (ANIM-09 grinding), clink (ANIM-02 ice)	—	—
hidden / idle 30 s	Gain → 0, then AudioContext.suspend()	800	linear

```
export function useAmbience() {
  const ctx = useRef<AudioContext>();
  const start = async () => {
    ctx.current ??= new AudioContext();
    const buf = await loadBuffer(ctx.current, mixFor(new
Date()));
    const src = ctx.current.createBufferSource();
    const gain = ctx.current.createGain();
    const an = ctx.current.createAnalyser(); an.fftSize = 256;
    src.buffer = buf; src.loop = true;

src.connect(gain).connect(an).connect(ctx.current.destination);
    gain.gain.setValueAtTime(0, ctx.current.currentTime);
    gain.gain.linearRampToValueAtTime(.35,
ctx.current.currentTime + 1.5);
    src.start();
    const data = new Uint8Array(an.fftSize); let lvl = 0;
    const tick = () => {
      an.getByteTimeDomainData(data);
      lvl += (rms(data) - lvl) * .15;
      document.documentElement.style.setProperty("--audio-level",
lvl.toFixed(3));
      raf = requestAnimationFrame(tick);
    };
    let raf = requestAnimationFrame(tick);
  };
  return { start, stop: () => ctx.current?.suspend() };
}
```

STATE MODEL

`off` → `loading` → `playing` → `ducked` → `suspended`

Ducks to 30 % while a video or screen reader speech is active (best-effort).

MOBILE / LOW-POWER

LITE: audio allowed, visual sync off (steam uses fixed loop).

REDUCED MOTION

Sound may play if user opts in, but no audio-reactive motion.

PERFORMANCE BUDGET

3 loops ≤ 300 KB total (Opus 48 kbps), lazy on toggle; analyser ≤ 0.3 ms/frame.

ACCEPTANCE CRITERIA

`ANIM-15-AC1` Audio never starts without an explicit user gesture.

`ANIM-15-AC2` Preference persists; a visible pause/volume control is always reachable (WCAG 1.4.2).

`ANIM-15-AC3` Visual sync stops when tab hidden; CPU idle within 1 s.

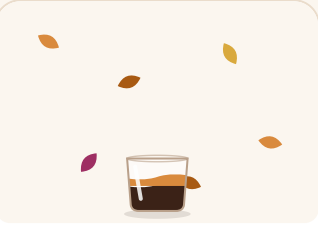
Seasonal particle themes

COULD | Canvas 2D (custom, 3 KB) | CMS season flag | OffscreenCanvas worker (FULL)

Trigger: **season flag from CMS (date-based default)**

CONCEPT

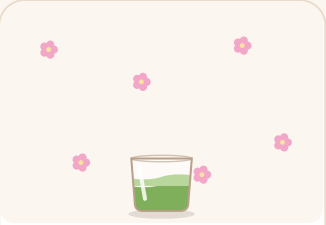
A lightweight particle layer behind the hero and seasonal banners changes with the season and campaign: autumn leaves tumbling, winter snow drifting, spring blossoms floating, summer nitro bubbles rising. Particles react subtly to scroll velocity (wind) and avoid the headline area.



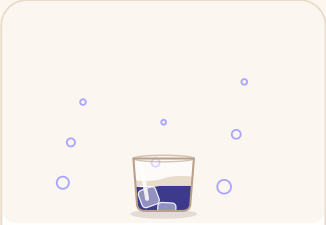
Sep–Nov Autumn: Pumpkin Spice leaves tumble.



Dec–Feb Winter: snow drifts; Berry Cold Brew.



Mar–May Spring: blossoms over Matcha Bloom.



Jun–Aug Summer: nitro fizz bubbles rise.

T / PHASE	KEYFRAME	DUR.	EASING
spawn	Density 18 (mobile) / 36 (desktop) particles	—	—
fall/rise	Velocity per theme (leaves 30–60 px/s, bubbles −40 px/s) + sine sway	loop	linear
scroll	Wind = scrollVelocity × 0.02, decays	—	lerp .1
season swap	Cross-fade old/new layer	800	standard

```
type Theme = "autumn" | "winter" | "spring" | "summer";
const CFG: Record<Theme, ParticleCfg> = {
  autumn: { n: 36, vy: [30, 60], spin: true, sprite: leafSprite
},
  winter: { n: 36, vy: [15, 35], spin: false, sprite: snowSprite
},
  spring: { n: 24, vy: [10, 25], spin: true, sprite:
blossomSprite },
  summer: { n: 30, vy: [-45, -25], spin: false, sprite:
bubbleSprite },
};
export function SeasonLayer({ theme }: { theme: Theme }) {
  const ref = useRef<HTMLCanvasElement>(null);
  const tier = useTier();
  useEffect(() => {
    if (tier === "static") return;
    const cfg = { ...CFG[theme], n: tier === "lite" ? 10 :
CFG[theme].n };
    const sim = createSim(ref.current!, cfg, { avoid: "#hero-
copy" });
    const io = new IntersectionObserver(([e]) =>
      e.isIntersecting ? sim.play() : sim.pause());
    io.observe(ref.current!);
    return () => { io.disconnect(); sim.destroy(); };
  }, [theme, tier]);
  return <canvas ref={ref} aria-hidden className="season-layer"
/>;
}
```

STATE MODEL

`off` → `loading-sprites` → `running` → `paused-offscreen`

Sprites are tiny SVG→bitmap atlases generated at runtime (no images downloaded).

MOBILE / LOW-POWER

LITE: 10 particles, 30 fps cap, no wind.

REDUCED MOTION

Static seasonal illustration (SVG) instead of particles.

PERFORMANCE BUDGET

≤ 36 particles; ≤ 1.5 ms/frame; zero network assets; paused off-screen.

ACCEPTANCE CRITERIA

`ANIM-16-AC1` Season resolves from CMS flag, else date (hemisphere setting).

`ANIM-16-AC2` Particles never overlap the headline bounding box.

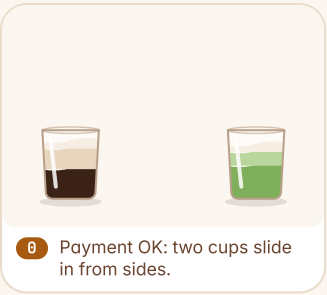
`ANIM-16-AC3` Removing the flag reverts to no particles without reload.

Checkout success “cheers”

SHOULD | Rive (state machine) or Lottie | Framer Motion (layout) | Web Audio clink (opt-in) | Trigger: **payment confirmed**

CONCEPT

The confirmation screen opens with a toast between two cups: the customer’s drink and a ‘barista’ cup slide in, tilt and clink with a tiny splash and sparkle, then the order headline, pickup code and ETA appear. A joyful, 1-second moment of closure.



`0` Payment OK: two cups slide in from sides.



`300 ms` Cups tilt toward each other.



`480 ms` Clink! splash droplets + sparkle lines.



`900 ms` Settle; headline + pickup code fade in.

T / PHASE	KEYFRAME	DUR.	EASING
0–300	Cups x ±60 → ±10, opacity 0→1	300	outExpo
300–480	Rotate ±16°, x → ±2 (contact)	180	accelerate
480–700	Splash particles, sparkle lines, liquid slosh	220	outQuart
700–1000	Cups settle 0°; headline y 12→0; code scale .9→1	300	spring.bouncy

```tsx
const Cheers = dynamic(() => import("@rive-app/react-canvas")
  .then(m => ({ default: (p: any) => <m.default {...p} /> })),
  { ssr: false });

export function SuccessHero({ order }: { order: Order }) {
  const tier = useTier();
  return (
    <section aria-labelledby="ok">
      {tier === "static"
        ? <img src="/motion/cheers-final.svg" alt="" />
        : <Cheers src="/motion/cheers.riv"
            stateMachines="success"
            layout={new Layout({ fit: Fit.Contain })} />}
      <motion.h1 id="ok" initial={{ opacity: 0, y: 12 }}
        animate={{ opacity: 1, y: 0 }}
        transition={{ delay: tier === "full" ? .7 : 0 }}>
        Order placed! Your {order.items[0].name} is next.
      </motion.h1>
      <PickupCode code={order.code} />
    </section>
  );
}
```

STATE MODEL

`paying` → `success` → `cheers` → `summary`

On payment failure: cups don’t meet; one cup wobbles “no” (ANIM-17-fail, 400 ms).

MOBILE / LOW-POWER

LITE: Lottie/Rive at 30 fps or simple 2-frame SVG tilt via CSS.

REDUCED MOTION

Final frame SVG; headline visible immediately.

PERFORMANCE BUDGET

Rive runtime ≤ 80 KB gz + .riv ≤ 40 KB, lazy-loaded during payment; ≤ 1 s duration.

ACCEPTANCE CRITERIA

`ANIM-17-AC1` Headline and pickup code are readable within 1 s (never blocked by animation).

`ANIM-17-AC2` Animation assets are preloaded while payment is processing.

`ANIM-17-AC3` Sound (if enabled) plays once and respects the global sound toggle.

Hold-to-confirm barista tamp

SHOULD | Framer Motion (useAnimate) | Pointer Events | Vibration API | Trigger: **press & hold the Place-order button (1.2 s)**

CONCEPT

Prevents accidental orders with a satisfying ritual: holding the Place-order button presses a tamper into a portafilter of grounds while a ring fills; releasing early springs it back. At 100 % the puck compresses with a haptic tick and the order is submitted.



0 Rest: tamper over loose grounds.



480 ms Holding: ring fills, tamper presses.



1.2 s 100 %: puck compressed, haptic.



+200 ms Confirmed → ANIM-17.

T / PHASE	KEYFRAME	DUR.	EASING
hold 0–1200	Ring 0→100 %, tamper y 0→16 px, grounds squash	1200	linear (time)
release < 100 %	Ring & tamper spring back	300	spring.snappy
100 %	Vibrate(12); puck scaleY .6; submit	120	outExpo
success	Morph to check, hand off to ANIM-17	200	standard

```
export function HoldToConfirm({ onConfirm, ms = 1200 }: Props) {
  const [scope, animate] = useAnimate();
  const ctrl = useRef<AnimationPlaybackControls>();
  const start = () => {
    ctrl.current = animate(scope.current, { "--p": 1 } as any,
      { duration: ms / 1000, ease: "linear",
        onComplete: () => { navigator.vibrate?.(12); onConfirm();
} });
  };
  const cancel = () => {
    ctrl.current?.stop();
    animate(scope.current, { "--p": 0 } as any, spring.snappy);
  };
  return (
    <button ref={scope} className="hold" aria-describedby="hold-
hint"
      onPointerDown={start} onPointerUp={cancel} onPointerLeave=
{cancel}
      onKeyDown={(e) => !e.repeat && (e.key === " " || e.key ===
"Enter") && start()}
      onKeyUp={cancel}>
      <TampSvg />  {/* reads var(--p) for ring + tamper */}
      <span>Hold to place order</span>
    </button>
  );
}
```

STATE MODEL

idle → holding → armed → submitting → done

Keyboard: hold Space/Enter; also a "Tap to confirm" setting for users who cannot hold.

MOBILE / LOW-POWER

LITE: ring only (no tamper art).

REDUCED MOTION

Hold replaced by standard tap + confirm dialog (setting also available to everyone).

PERFORMANCE BUDGET

CSS var drives SVG; ≤ 0.5 ms/frame; no network until confirmed.

ACCEPTANCE CRITERIA

ANIM-18-AC1 Releasing before 1.2 s never submits.

ANIM-18-AC2 Accessible alternative (tap + confirm) available and announced (WCAG 2.5.1 / 2.5.7).

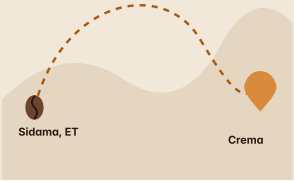
ANIM-18-AC3 Exactly one submission per completed hold (debounced).

Origin map bean flight

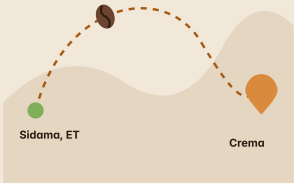
COULD · GSAP MotionPathPlugin · ScrollTrigger (scrub) · SVG map · Trigger: **About page chapter scrolls into view**

CONCEPT

On the Our Story page, a roasted bean travels from the farm (e.g. Sidama, Ethiopia) along a dotted route to the roastery and finally the café, scrubbed by scroll. Each waypoint reveals a short story card with photo, altitude, process and tasting notes.



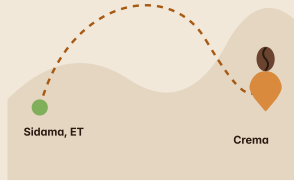
0 % Origin pin pulses; bean appears at farm.



45 % Bean follows dotted route, rotating.



90 % Passes roastery waypoint (label pops).



100 % Lands on café pin; story chapter fades in.

T / PHASE	KEYFRAME	DUR.	EASING
0–0.1	Origin pin ring pulse ×2	—	sine
0.1–0.9	Bean along path (motionPath, autoRotate)	scrub	none
0.45	Roastery card slides in	0.08	power2.out
0.9–1	Café pin bounce; next chapter fade	0.1	back.out(2)

```
gsap.registerPlugin(ScrollTrigger, MotionPathPlugin);
useGSAP(() => {
  const tl = gsap.timeline({ scrollTrigger: {
    trigger: ".origin-map", start: "top 70%", end: "bottom 30%",
    scrub: .8 } });
  tl.to(".origin-pin .pulse", { scale: 2, autoAlpha: 0,
       repeat: 1, duration: .1 })
    .to(".bean", { duration: .8, ease: "none",
       motionPath: { path: "#route", align: "#route",
                     autoRotate: true, alignOrigin: [.5, .5] }
})
    .from(".card-roastery", { x: 40, autoAlpha: 0,
       duration: .08 }, .45)
    .from(".cafe-pin", { y: -12, ease: "back.out(2)",
       duration: .1 }, .9);
}, { scope: root });
```

STATE MODEL

`off` → `origin` → `travel` → `roastery` → `arrived`

Each bean origin is data-driven (JSON of waypoints).

MOBILE / LOW-POWER

LITE: time-based 2 s play when in view (no scrub).

REDUCED MOTION

Static map with full route drawn and all cards visible.

PERFORMANCE BUDGET

MotionPath plugin ≤ 10 KB gz; SVG map ≤ 25 KB; shares GSAP chunk with ANIM-01.

ACCEPTANCE CRITERIA

`ANIM-19-AC1` Bean stays exactly on the route at all scroll positions.

`ANIM-19-AC2` Route and cards are readable as a list for screen readers.

`ANIM-19-AC3` Works with data for any origin (2–5 waypoints).

Gyro tilt liquid

COULD | DeviceOrientation API | Custom spring (shared with ANIM-02) | SVG | Trigger: **device tilt on mobile (after permission)**

CONCEPT

On phones, tilting the device makes the coffee in the hero and customizer cups stay level with the real world, sloshing slightly with a spring. A small, delightful proof that the cup is 'real'.



0° Phone upright: level surface.



-18° Tilt left: liquid stays level with world.



+22° Tilt right: slosh overshoots then settles.



perm iOS: permission prompt via a button.

T / PHASE	KEYFRAME	DUR.	EASING
orientation	Target angle = −gamma, clamp ±30°	live	—
spring	Surface angle follows target (k 120, c 10)	~600 settle	spring
fast tilt	Slosh amplitude ∝ angular velocity, max 5 px	—	—
permission	iOS requires DeviceOrientationEvent.requestPermission() on tap	—	—

```
export function useTiltAngle() {
  const target = useRef(0);
  const angle = useSpring(0, { stiffness: 120,
damping: 10 });
  const enable = async () => {
    const D = DeviceOrientationEvent as any;
    if (typeof D?.requestPermission ===
"function") {
      if ((await D.requestPermission()) !==
"granted") return false;
    }
    addEventListener("deviceorientation", (e) =>
{
      target.current = Math.max(-30, Math.min(30,
-(e.gamma ?? 0)));
      angle.set(target.current);
    }, { passive: true });
    return true;
  };
  return { angle, enable };   // <motion.g style=
{{ rotate: angle }}>
}
```

STATE MODEL

`unsupported` → `needs-permission` → `active` → `paused`

Paused when cup off-screen or page hidden.

MOBILE / LOW-POWER

Disabled in LITE (sensor events cost battery).

REDUCED MOTION

Disabled; no permission prompt shown.

PERFORMANCE BUDGET

Sensor events throttled to rAF; ≤ 0.5 ms/frame.

ACCEPTANCE CRITERIA

`ANIM-20-AC1` Never requests permission without a user tap on an explanatory button.

`ANIM-20-AC2` Liquid stays level within ±3° after settling.

`ANIM-20-AC3` Feature silently absent on desktop / unsupported devices.

Foam-text headline reveal

COULD | SVG mask / clip-path | Framer Motion whileInView | CSS | Trigger: **section headline enters viewport**

CONCEPT

Section headlines on editorial pages (Story, Seasonal) are 'poured' in milk foam: a bubbly foam front sweeps across a crema panel and leaves the letters behind in microfoam, like latte-art lettering.



0 Crema panel; text hidden.



300 ms Foam front sweeps right, bubbles at edge.



600 ms Letters appear as microfoam.



800 ms Settled; subtle bubble texture.

T / PHASE	KEYFRAME	DUR.	EASING
0–800	Mask rect width 0→100 %	800	pour
0–800	Bubble front follows edge, jitter ±4 px	800	—
600–900	Bubbles pop (scale→0)	300	outQuart
after	Static, real text underneath for SEO/AT	—	—

```
export function FoamHeadline({ children }: { children: string })
{
  return (
    <h2 className="foam-h">
      <span className="sr-only">{children}</span>
      <svg aria-hidden viewBox="0 0 600 120">
        <mask id="m"><motion.rect height={120} fill="#fff"
          initial={{ width: 0 }} whileInView={{ width: 600 }}
          viewport={{ once: true, amount: .6 }}
          transition={{ duration: .8, ease: [.65,0,.35,1] }} />
</mask>
        <text mask="url(#m)" className="foam-text">{children}
</text>
        <FoamFront />
      </svg>
    </h2>
  );
}
```

STATE MODEL

`hidden` → `pouring` → `settled`

Runs once per session per headline.

MOBILE / LOW-POWER

LITE: mask reveal without bubbles.

REDUCED MOTION

Text simply visible (no reveal).

PERFORMANCE BUDGET

SVG only, ≤ 2 KB; once per headline.

ACCEPTANCE CRITERIA

`ANIM-21-AC1` Real heading text exists in DOM for SEO and screen readers.

`ANIM-21-AC2` Reveal fires once at 60 % visibility.

`ANIM-21-AC3` Foam text contrast on crema ≥ 4.5:1 (#F7EEE3 on #8C5A33 panel variant).

Receipt print-out confirmation

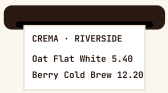
SHOULD | Framer Motion | CSS steps() for print jitter | Trigger: **order confirmed (after ANIM-17) or "View receipt"**

CONCEPT

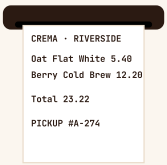
The digital receipt prints out of a slim slot like a café ticket: it feeds in small jittery steps, lines appear as they 'print', the QR and pickup code land last, and a perforated tear edge appears. Swiping it down saves to Wallet / downloads PDF.



0 Confirmed: printer slot appears.



400 ms Receipt feeds out line by line.



900 ms QR + pickup code print.



1.1 s Tear edge; drag down to save to Wallet.

T / PHASE	KEYFRAME	DUR.	EASING
0–150	Slot fades in	150	outQuart
150–1000	Receipt y −100 % → 0 in 12 steps (steps(12)) + 1 px x-jitter	850	steps
each line	Line opacity 0→1 as it passes slot	60	linear
1000–1100	Tear edge draws; drag-to-save hint	100	outQuart

```
export function PrintedReceipt({ order }: { order: Order }) {
  const tier = useTier();
  const lines = receiptLines(order);
  return (
    <div className="slot" aria-label="Receipt">
      <motion.article className="receipt"
        initial={tier === "static" ? false : { y: "-100%" }}
        animate={{ y: 0 }}
        transition={{ duration: .85, ease: steps(12) }}
        drag="y" dragConstraints={{ top: 0, bottom: 0 }}
        onDragEnd={(_, i) => i.offset.y > 80 &&
saveToWallet(order)}>
        {lines.map((l, i) => (
          <motion.p key={i} initial={{ opacity: 0 }} animate={{
opacity: 1 }}
            transition={{ delay: .15 + i * .07 }}>{l}</motion.p>
        ))}
        <PickupQR code={order.code} />
      </motion.article>
    </div>
  );
}
const steps = (n: number) => (t: number) => Math.floor(t * n) /
n;
```

STATE MODEL

hidden → printing → printed → saving → saved

Re-opening the receipt later skips printing (static).

MOBILE / LOW-POWER

LITE: smooth 400 ms slide (no steps/jitter).

REDUCED MOTION

Receipt shown immediately; save buttons visible.

PERFORMANCE BUDGET

DOM only, ≤ 3 KB; QR generated client-side ≤ 6 KB lib.

ACCEPTANCE CRITERIA

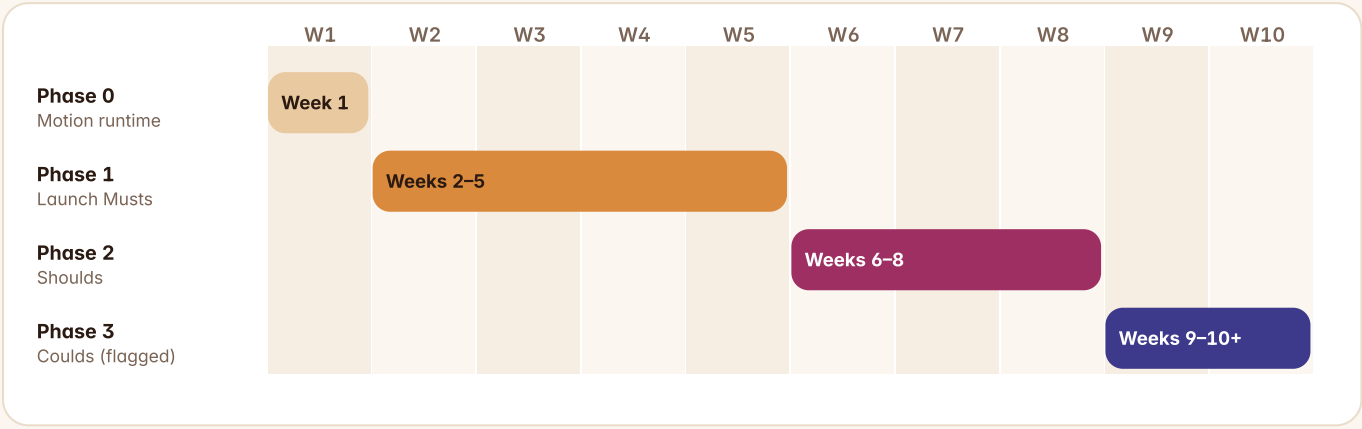
ANIM-22-AC1 Full receipt content is accessible as text (not image).

ANIM-22-AC2 Save-to-Wallet and Download PDF buttons exist besides the drag gesture.

ANIM-22-AC3 Printing completes ≤ 1.1 s and never blocks the pickup code display.

From foundations to delight

Four phases over ~10 weeks, aligned with the UI spec design timeline. Must items gate launch; Could items ship behind flags after launch metrics are healthy.



PHASE 0 · WEEK 1 · MOTION RUNTIME

MotionProvider, tier detection, FPS guard, tokens, Storybook tier toggles, Playwright motion harness

PHASE 1 · WEEKS 2–5 · LAUNCH MUSTS

ANIM-01 hero pour, ANIM-02 live cup, ANIM-04 add-to-cart, ANIM-05 loader, ANIM-09 tracking

ANIM-01 · ANIM-02 · ANIM-04 · ANIM-05 · ANIM-09

PHASE 2 · WEEKS 6–8 · SHOULDS

ANIM-03, 08, 10, 12, 13, 17, 18, 22

ANIM-03 · ANIM-08 · ANIM-10 · ANIM-12 · ANIM-13 · ANIM-17 · ANIM-18 · ANIM-22

PHASE 3 · WEEKS 9–10+ · COULDS (FLAGGED)

ANIM-06, 07, 11, 14, 15, 16, 19, 20, 21

ANIM-06 · ANIM-07 · ANIM-11 · ANIM-14 · ANIM-15 · ANIM-16 · ANIM-19 · ANIM-20 · ANIM-21

Definition of done (per animation)

GATE	REQUIREMENT
Spec parity	Frames match storyboard key frames (Playwright golden screenshots per tier)
Three tiers	FULL, LITE and STATIC versions implemented and visible in Storybook
Budgets	Per-spec JS/asset/frame budgets verified in CI (size-limit + trace)
Accessibility	Reduced-motion verified; no information conveyed only by motion; live-region text present
Flags	Could items wrapped in `flags.anim.<id>`; kill-switch tested

Dependencies & risks

ITEM	NOTE
ANIM-02 → 07, 09, 17	Layer model (`buildLayers`) must be stable before 3D, tracking and cheers reuse it.
GSAP chunk	ANIM-01 and ANIM-19 share one lazy GSAP chunk; never load GSAP on routes without them.
iOS scroll pinning	Address-bar resize can jitter pinned hero; mitigated by LITE time-based play on small screens.

07 · ASSET LIST

Assets to produce

Owners: **Design** produces SVG/Lottie/Rive/3D; **Frontend** optimises (SVGO, dotLottie, gltf-transform) and enforces budgets via size-limit in CI.

TYPE	ASSET	USED BY	BUDGET	SPEC / NOTES
SVG	`portafilter.svg, kettle.svg`	ANIM-01	≤ 6 KB each	Inline, ids per part (spout, body, stream-anchor)
SVG	`cup-glass.svg (back/front/clip)`	01, 02, 09, 17	≤ 5 KB	Shared clipPath id, 120×160 viewBox
SVG	latte-art-heart.svg, rosetta.svg, leaf.svg	02, 05	≤ 2 KB	Single stroke paths, pathLength=1
SVG	`bean.svg, bean-sprite atlas`	03, 13, 19	≤ 2 KB	Used for particles via drawImage
SVG	`tamper-portafilter.svg`	ANIM-18	≤ 4 KB	Reads CSS var --p
SVG	`origin-map.svg (route path)`	ANIM-19	≤ 25 KB	#route path, waypoint groups
SVG	season sprites (leaf, snow, blossom, bubble)	ANIM-16	≤ 3 KB	Rasterised at runtime
Lottie (.lottie)	`bean-burst.lottie`	09, 13	≤ 60 KB	dotLottie, 60 fps, ≤ 1.2 s, no images
Lottie (.lottie)	`milk-pour.lottie (optional)`	ANIM-02	≤ 60 KB	Alternative to SVG swirl
Rive (.riv)	cheers.riv (state machine: success/fail)	ANIM-17	≤ 40 KB	Inputs: success:boolean, fail:trigger
Rive (.riv)	`stamp.riv (optional)`	ANIM-13	≤ 30 KB	Input: earn:trigger, seed:number
3D (.glb)	`cup.glb (glass + liquid stack)`	ANIM-07	≤ 500 KB	Draco, 1 material glass, ≤ 15k tris
Env map	`studio-warm.hdr → .ktx2`	ANIM-07	≤ 150 KB	1k, warm café lighting
Image	cup-poster.avif, cup-sprite-8.avif	01, 07	≤ 60 KB each	STATIC/LITE fallbacks
Audio	cafe-morning.opus, cafe-night.opus, rain.opus	ANIM-15	≤ 300 KB total	48 kbps, seamless loops
Audio	clink.opus, grinder.opus, thunk.opus	02, 09, 13, 17	≤ 15 KB each	One-shots, −14 LUFS

EXPORT CHECKLIST

SVG: SVGO, no embedded raster, ids kebab-case, viewBox kept. Lottie: no expressions, no images, ≤ 60 fps. Rive: named state machine + inputs documented.

COLOUR

All assets use token hex values from UI spec v1.0; dark variants via CSS currentColor / Rive colour inputs, never duplicated files.

TOTAL BUDGET

All motion assets on any single route ≤ 400 KB (excluding optional 3D), loaded lazily after LCP.

Testing motion like code

Run before each phase gate. Devices: iPhone 13 (Safari), Pixel 6a (Chrome), mid Windows laptop (Edge), MacBook (Safari/Chrome/Firefox).

CHECK	FULL	LITE	STATIC
Frames match storyboard (golden screenshots)	✓	✓	✓
60 fps (desktop) / ≥ 50 fps (mid phone), 10 s trace	✓	≥ 45	—
No long tasks > 50 ms during animation	✓	✓	✓
Heavy libs not requested (GSAP, three, Rive) when unused	—	✓	✓
Loops pause off-screen & on hidden tab	✓	✓	—
aria-live / text alternative present	✓	✓	✓
Keyboard & screen-reader path complete	✓	✓	✓
Interrupt / rapid input safe (no stuck states)	✓	✓	✓
Bundle and asset budgets (size-limit)	✓	✓	✓

Reduced-motion summary

ID	STATIC BEHAVIOUR
ANIM-01	Static full-cup poster, no pin
ANIM-02	Instant recipe state, 150 ms fade
ANIM-03	150 ms page cross-fade
ANIM-04	Badge update + toast
ANIM-05	Static heart + text
ANIM-06/11/16/20	Disabled
ANIM-07	Poster image
ANIM-08	Colour change on press
ANIM-09	Instant step updates

ID	STATIC BEHAVIOUR
ANIM-10	Instant theme swap
ANIM-12	Indicator jumps
ANIM-13	Ring fade-in
ANIM-14	Refresh button + loader
ANIM-15	Audio only, no visual sync
ANIM-17	Final frame
ANIM-18	Tap + confirm dialog
ANIM-19	Static route map
ANIM-21/22	Static text / receipt

SIGN-OFF

Owner	Frontend lead	Motion designer
Tanvir Hasan		
Signature · date	Signature · date	Signature · date